

Autonomous Truck Deployment – Impact of Road Construction

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Presented by: **Salah Uddin Momtaz**

Presented to: **Aurora**



Journey Through the Data



Setting the Stage: AV Trucks Meet Reality



Uncovering the Hidden Patterns



Emerging City Construction Landscape



Cities to Routes: Tires on Pavement



Predicting Construction Chaos



Analysis to Action



Discussion: Path Forward

Setting the Stage: Where AV Trucks Meet Reality

Promise vs. Daily Roadblocks

\$168B

*Annual savings promised
by autonomous trucks*

8.8K

*Construction causing
daily disruptions in 2021*

*Autonomous fleet needs to **navigate this maze of construction** –
minimizing costly **delays, route failures**, and customer **dissatisfaction***

Data Sources Behind the Story



*US Road Construction and
Closures (2016 - 2021)*



Route Line



Census Data



Road Network GIS

Uncovering the Hidden Patterns

What 6.2 Million Construction Projects Revealed

Goals

- Data analytics: US road construction data (2016-2021)
- Critical insights for AV deployment strategy
- Identifying high-impact zones and routes
- Predict the impact of constructions



22.6K
Total Cities Analyzed



5
Self-driving Routes



6.2M
Construction Projects



2.7B
Construction Hours

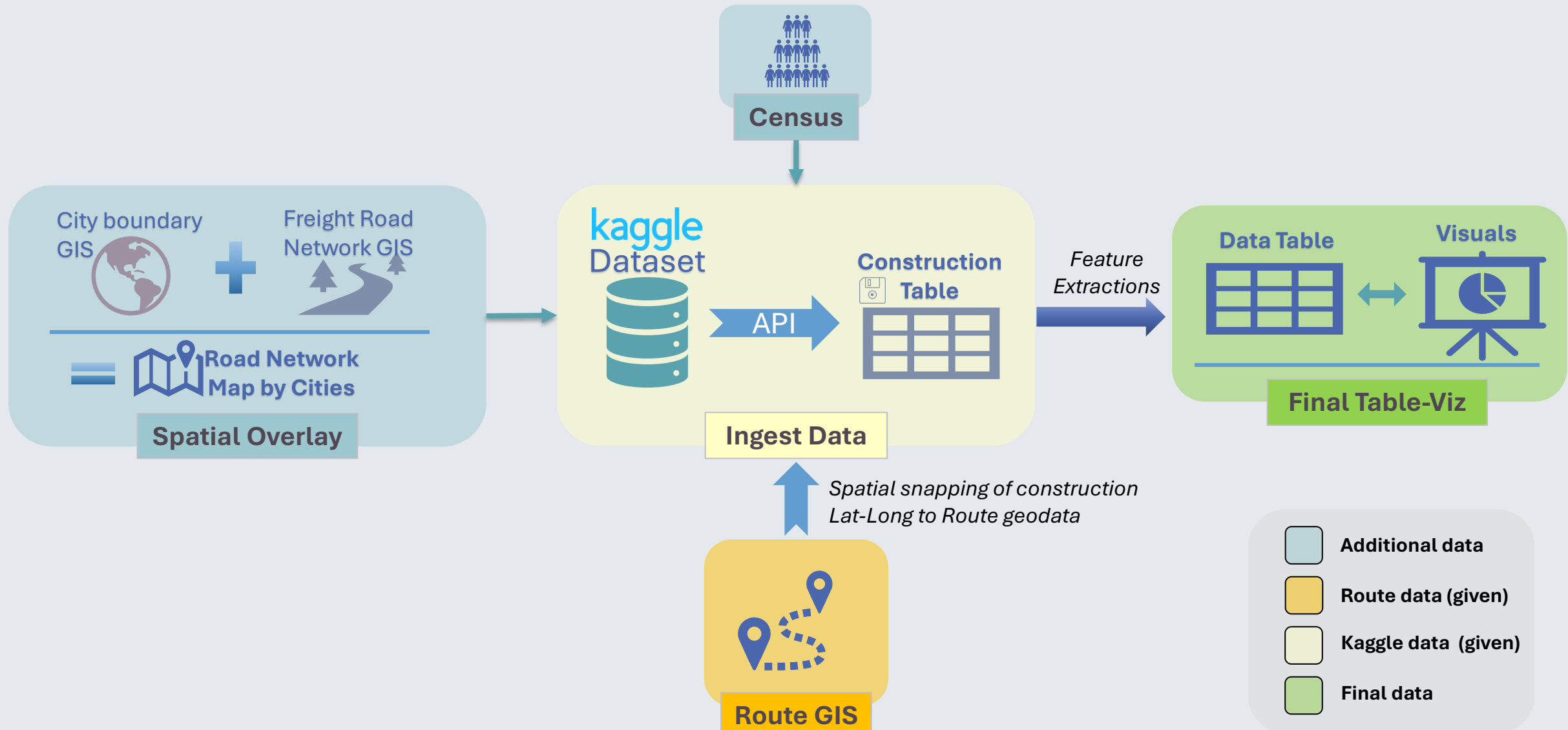


7.8M
Construction Length



6Yr
Continuous Data

Connecting Construction - Route

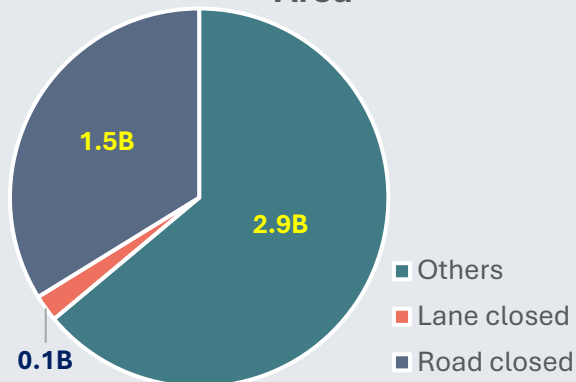


First Discovery: Not All Disruptions Are Equal

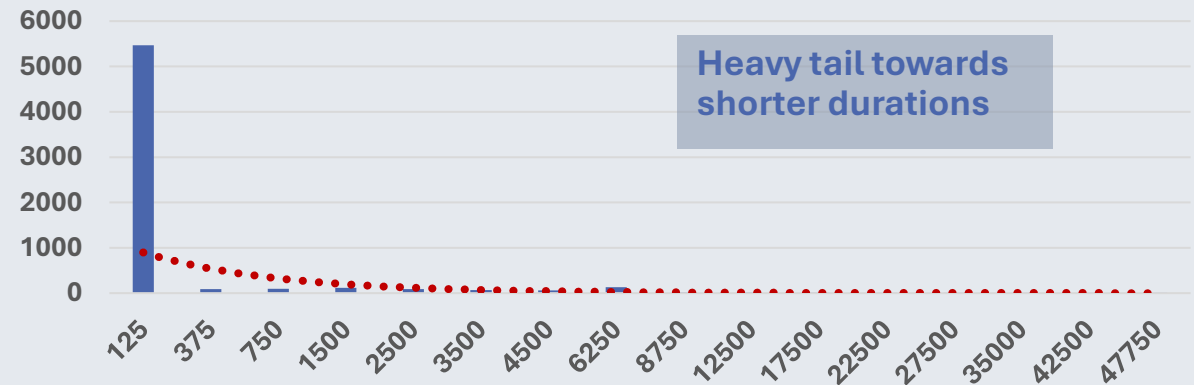
Summary Steps

- Feature Extractions
 - Duration → From start and end time
 - Road/Lane Closure → From “Description”
- Distribution & Missing Values

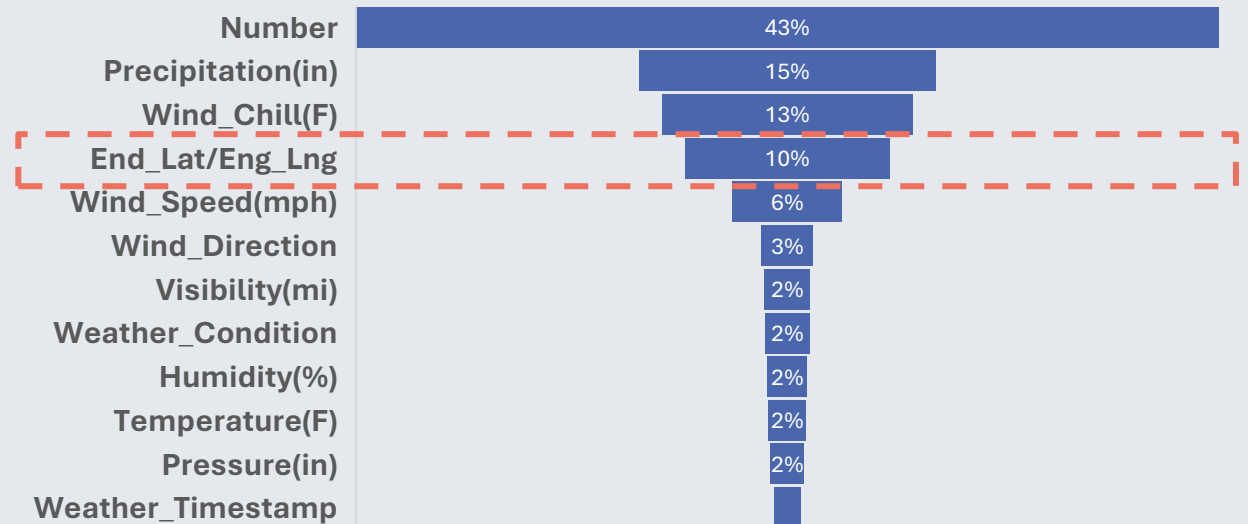
Duration (hour) by
Construction Type – Bay
Area



Duration Histogram



% Missing



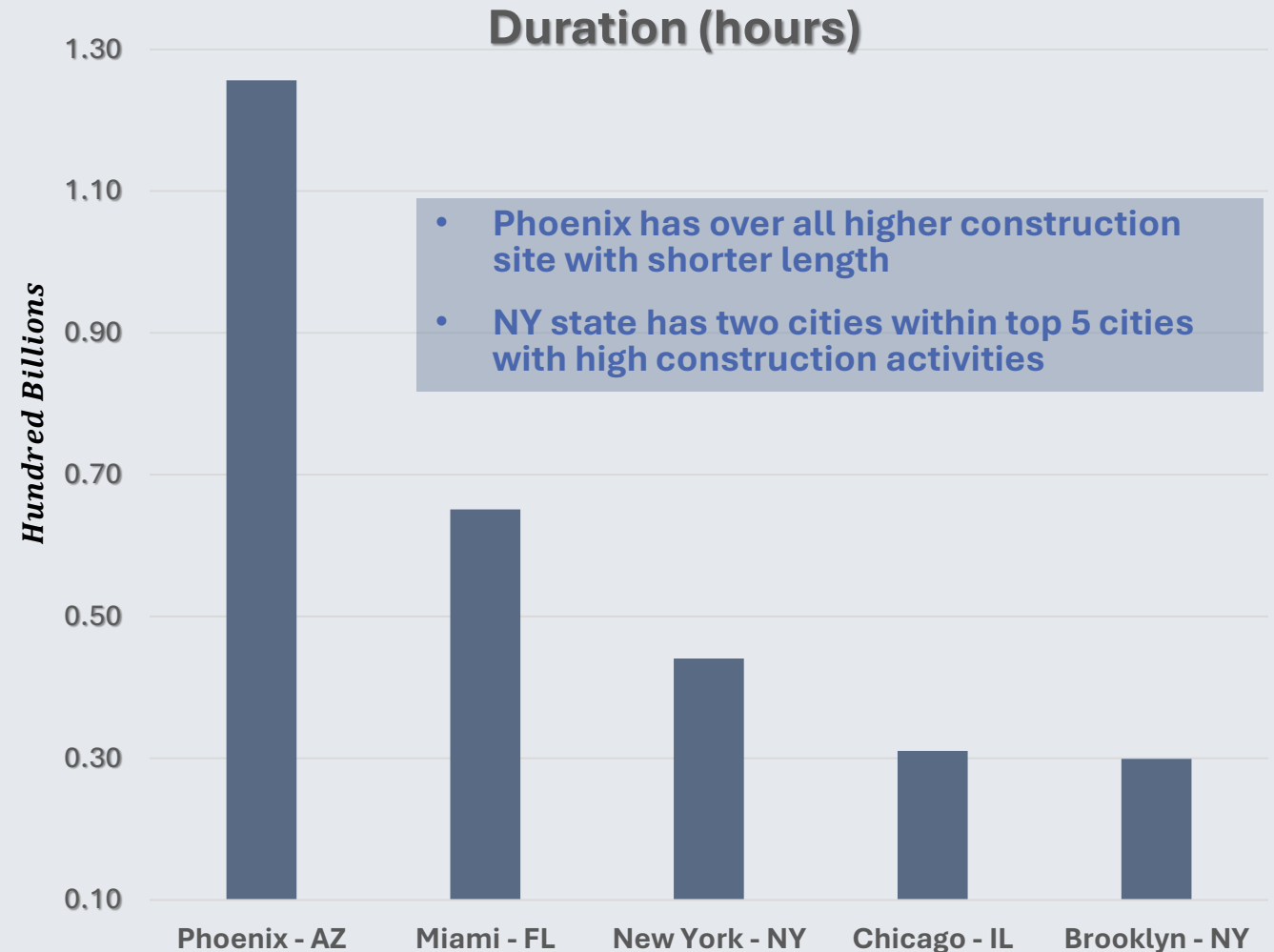
The City Construction Landscape Emerges

Phoenix Paradox: Most Construction, Shortest Delays

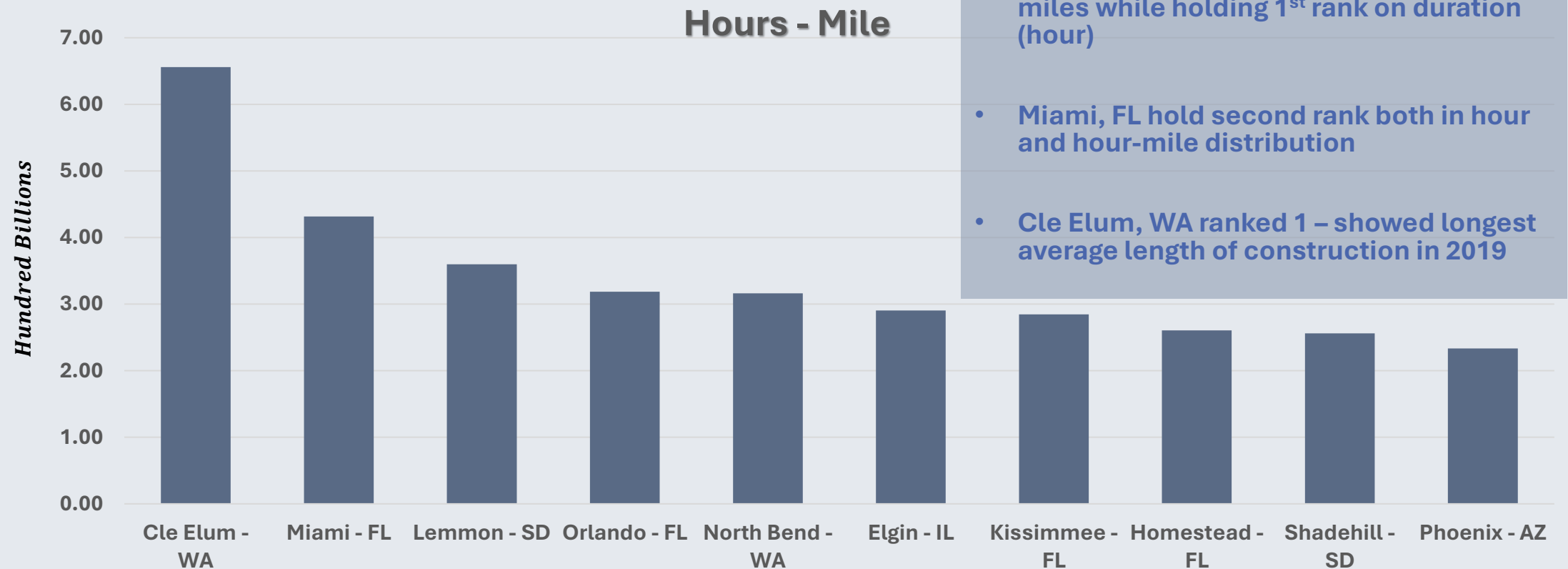
Top Cities Over 6 years

Year	 Top City	Million Hours	Sites	Avg. Distance (miles)
2016	Middletown, DE	10.4	299	0.9
2017	York, PA	10.3	841	1.8
2018	Winnfield, LA	9.8	461	1.9
2019	Cle Elum, WA	17.8	1,393	4.0
2020	Phoenix, AZ	11.9	34,240	0.2
2021	Phoenix, AZ	10.3	212,356	0.2

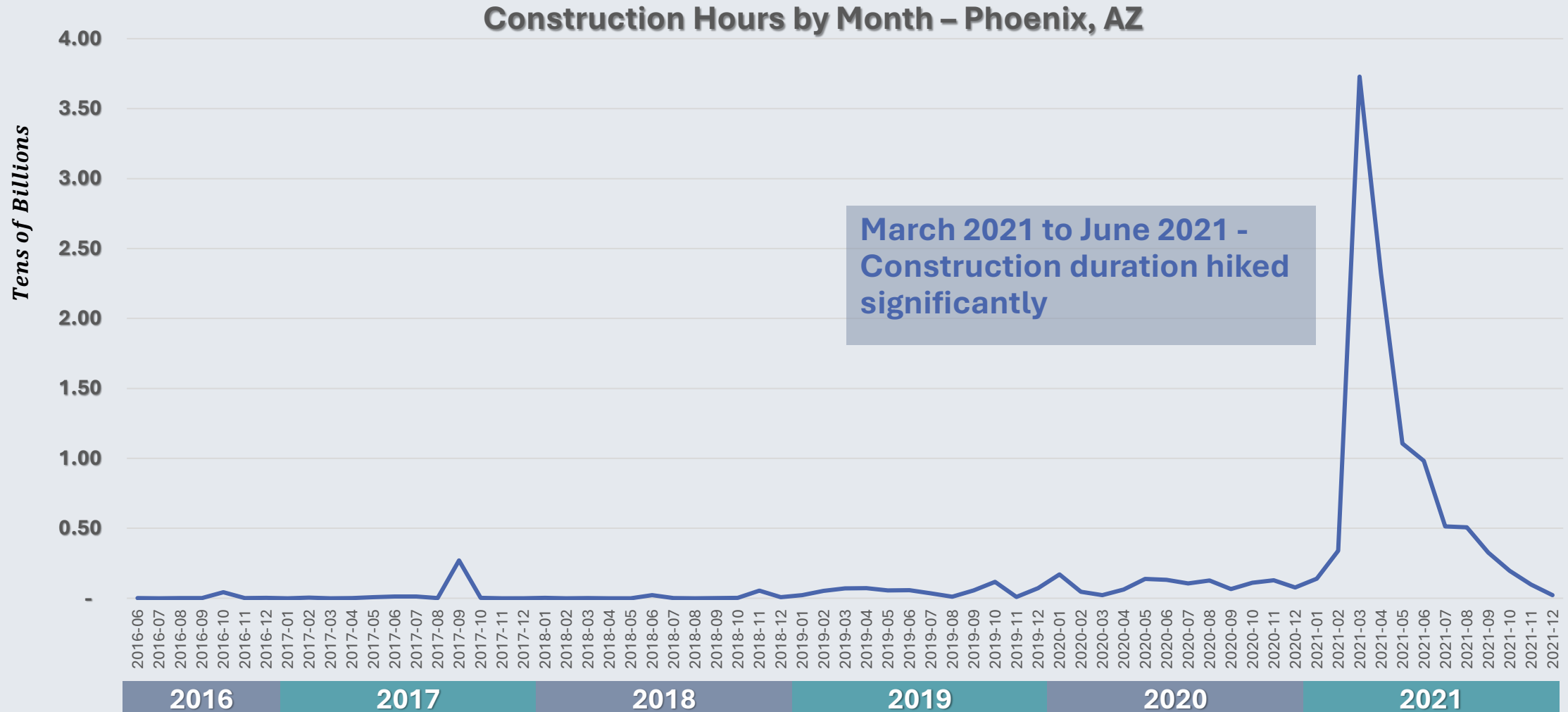
- Phoenix has higher number of construction sites with shorter length
- Cle Elum has long stretched sites



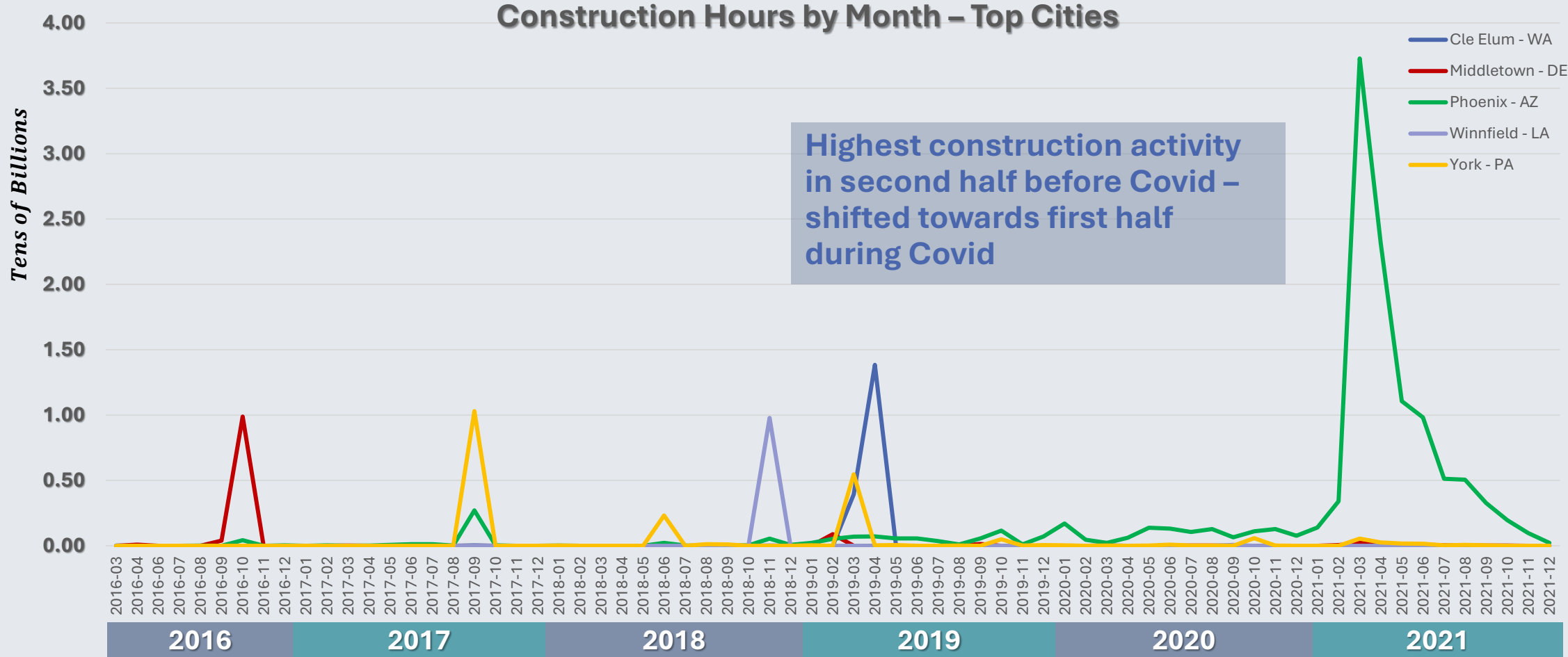
More Phoenix Paradox: Ranks #1 in Hours but #10 in Impact



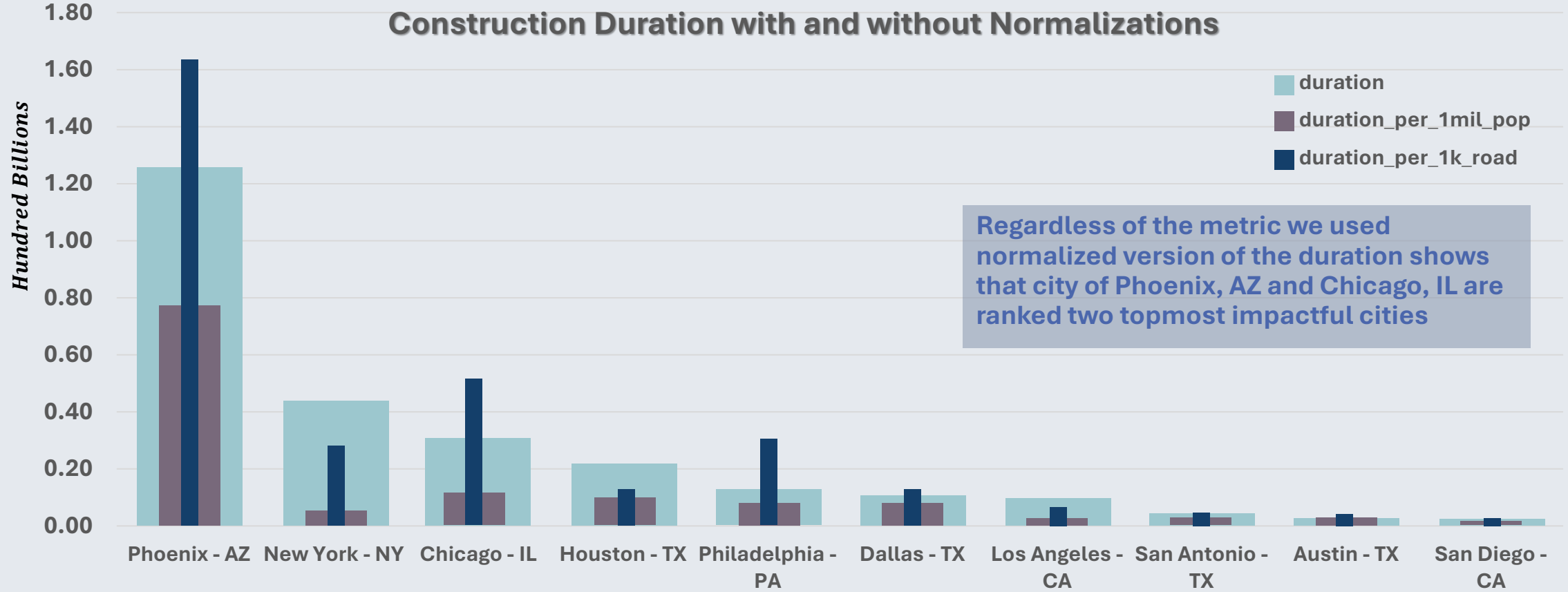
March 2021: Construction Patterns Change



COVID's Gift: Shifting Construction Seasons

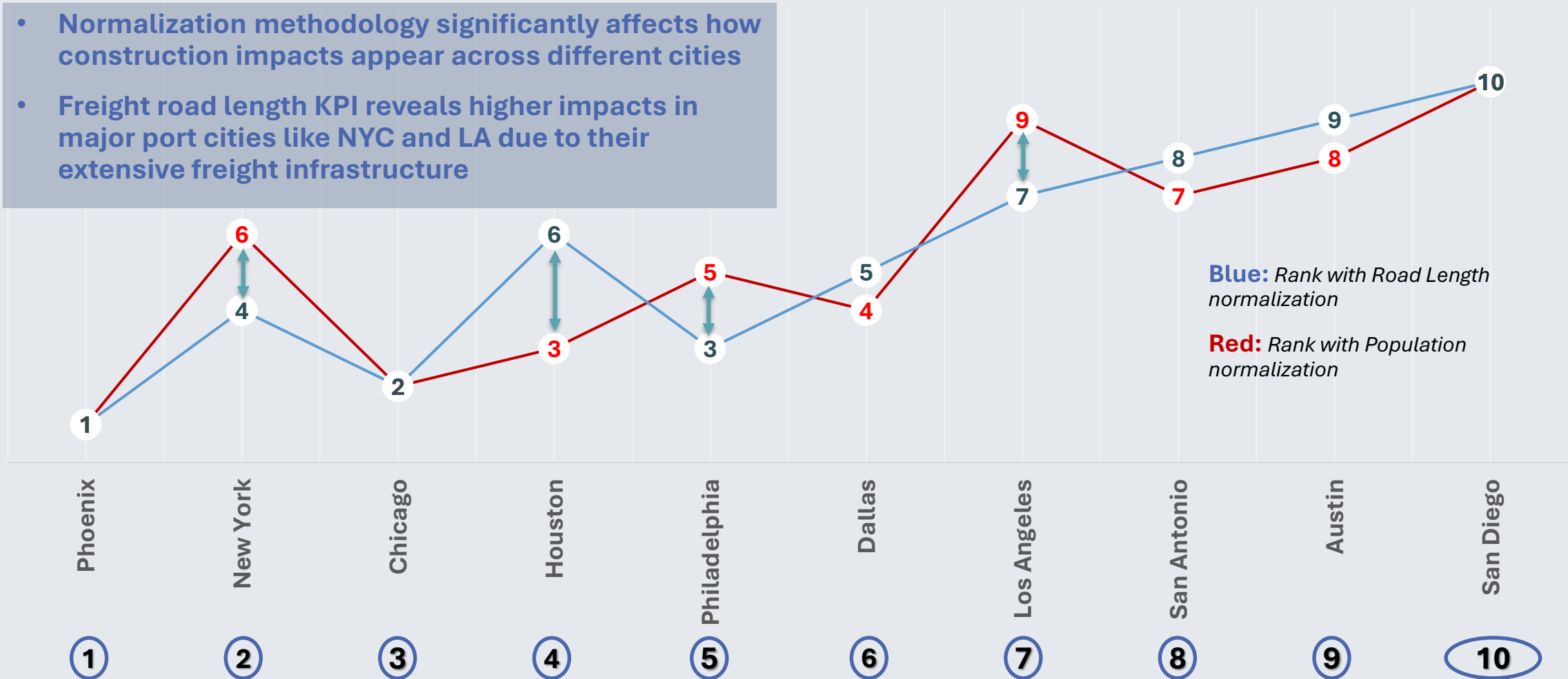


Phoenix & Chicago: The Construction Giants



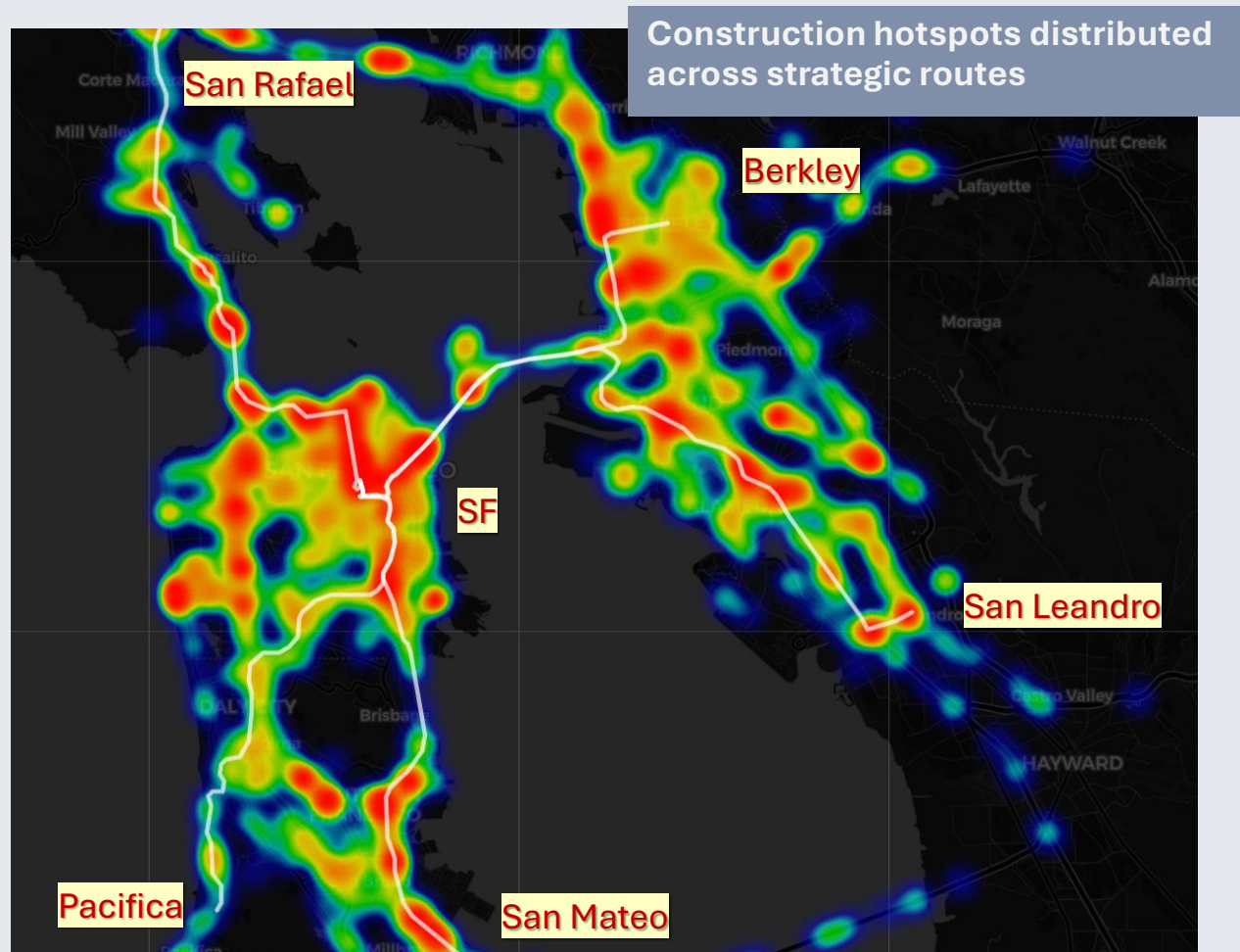
Normalization Method Changes Everything

- Normalization methodology significantly affects how construction impacts appear across different cities
- Freight road length KPI reveals higher impacts in major port cities like NYC and LA due to their extensive freight infrastructure



From Cities to Routes: Where Rubber Meets Road

Heat Map Reveals: One Route Stands Apart



Geospatial Analysis Summary

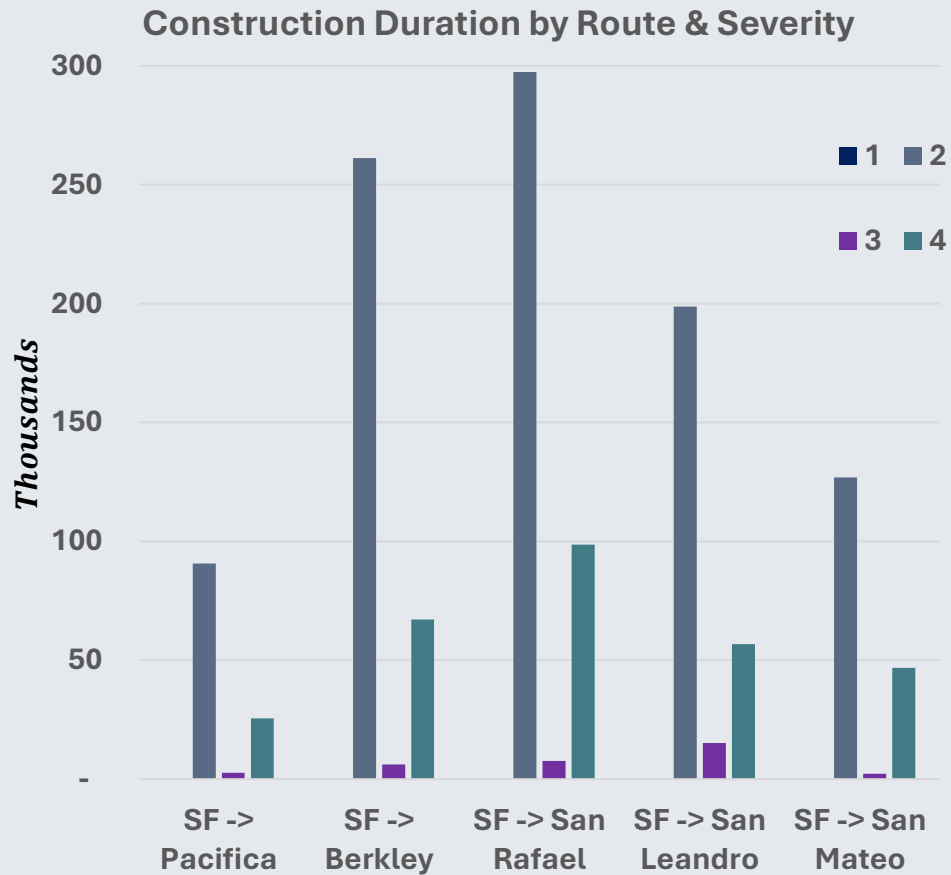
- 32K+ total construction events in Bay Area
- 20m radius threshold for spatial snapping to matching construction sites with Routes
- 9K+ events directly impacting routes

SF → Pacifica shows visibly lower construction density

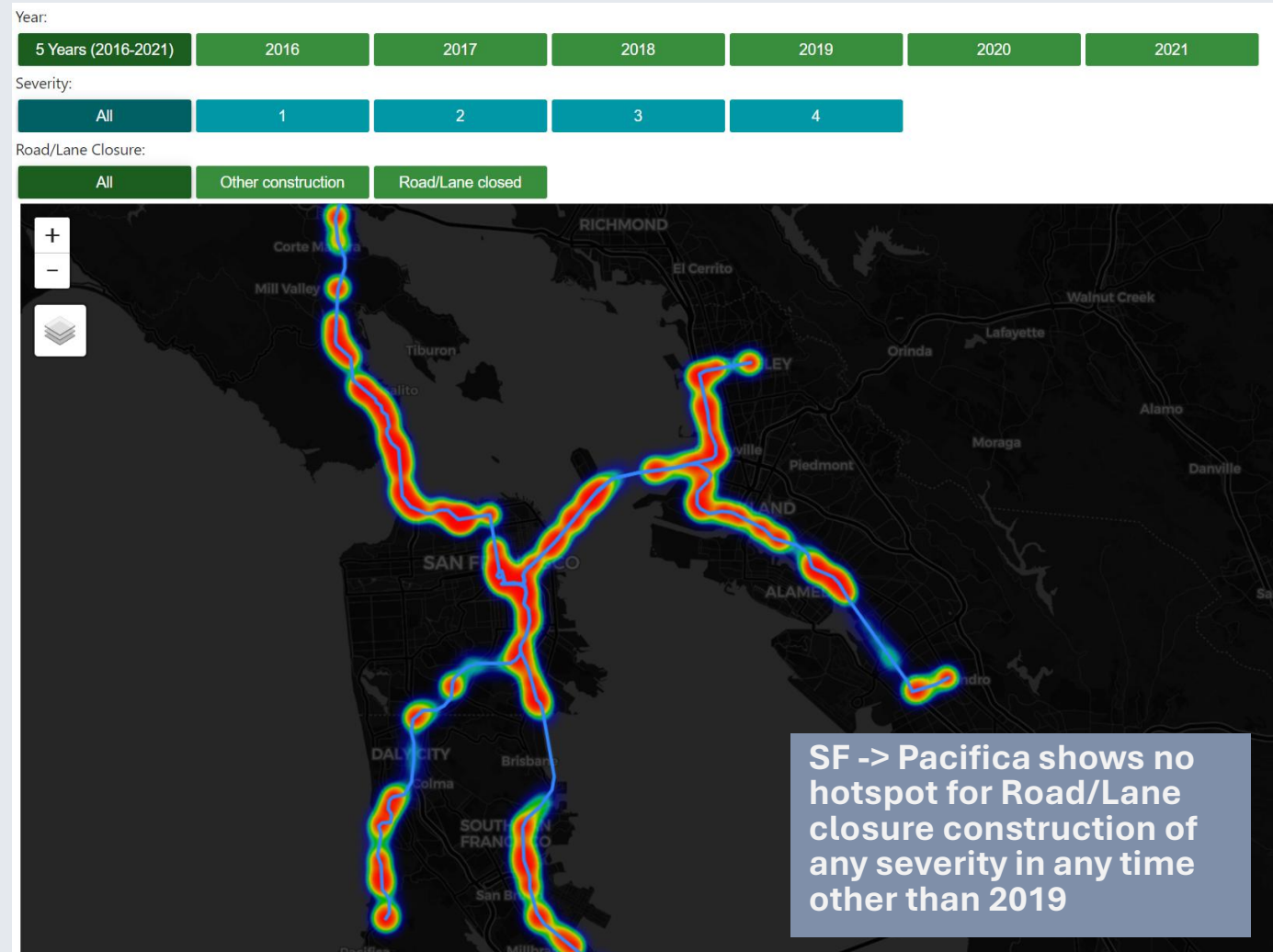
SF → San Rafael traverses high-intensity zones

Construction hotspots **concentrate around urban core**

SF-Pacifica: The Clear Winner for AV Testing



SF -> Pacifica shows lowest number of construction with higher severity



Quantifying Risk: From Best (1) to Worst (0)

Risk Scoring Methodology

Feature **Weights** (by Standard Deviation)

 Construction Duration: **96.5%**

 Distance Impact: **1.3%**

 Severity Weight: **1.4%**

 Road Closure Events: **0.8%**

Risk Score – Developed by Inverse Normalization

Route	Risk Score	Rank	Hours	Road Closure	Distance (mi)
SF → Pacifica	1.00	1	119K	1.1K	0.9K
SF → San Mateo	0.80	2	176K	1.6K	1.2K
SF → San Leandro	0.45	3	271K	3.1K	4.2K
SF → Berkeley	0.24	4	335K	3.2K	3.9K
SF → San Rafael	0.01	5	404K	2.9K	2.7K

Can We Predict Construction Chaos?

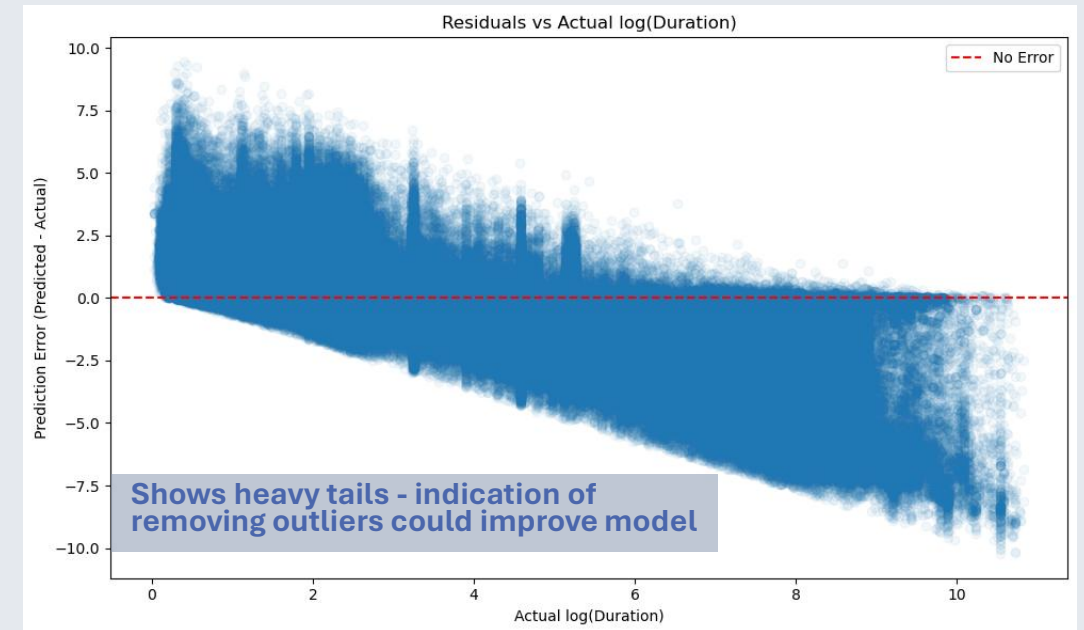
Random Forest Wins

Log Linear Regression Model

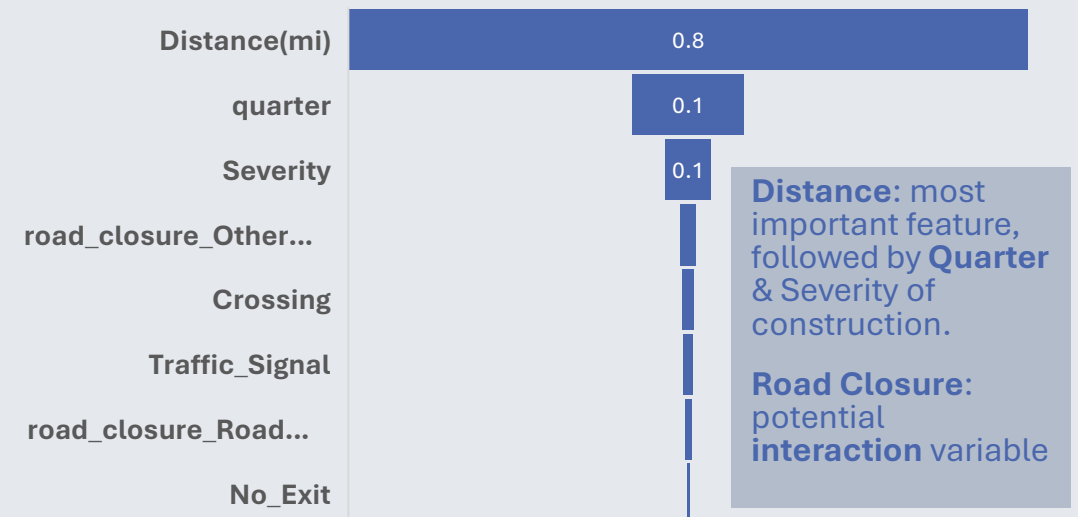
- Training, Test data split 80% - 20%
- Performs poorly
 - MAE: 1.645
 - RMSE: 2.132
 - R^2 : 0.076

Random Forest Regressor

- Training, Test data split 80% - 20%
- **Good** enough for strategic/sketch planning, needs improvement for daily operations
 - MAE: 1.2344
 - RMSE: 1.851
 - R^2 : 0.303



Feature Importance - Random Forest



The Path to Improve Predictability

Current Performance Benchmark: 30% R^2 - insights for strategic decisions

Target Enhancement: 50%+ R^2 - could aid operational automations and dynamic deployments

Model Enhancement Roadmap

Outlier Treatment

- *Remove top or bottom 5%*
- *Down-weight the outliers*

Interaction Terms

- *Road closure, Severity, Quarters and Distance interactions could work*

Ensemble Methods

- *Evaluate Geospatial Models*
- *Segmented model for different spatial levels*

Models Helps to Quantify Actions

Predict Duration → Estimate Delay → Estimate Operating Cost → Score by Cost → **Zore-cone – Minimal cost Route**

Route	Avg Duration (Predicted)	Annual Events	Delay Factor	Exposure Factor	Est. Delay (hours)	Operating Cost	Rank by Cost
SF → Pacifica	43.7	240	0.28	0.28	820	\$125K	1
SF → San Mateo	38.2	341	0.29	0.43	1,620	\$245K	2
SF → San Leandro	47.0	603	0.22	0.44	2,640	\$410K	4
SF → Berkeley	43.8	652	0.20	0.50	2,860	\$430K	5
SF → San Rafael	44.7	618	0.22	0.34	2,070	\$310K	3

*Saves
\$300K
per truck
per year*

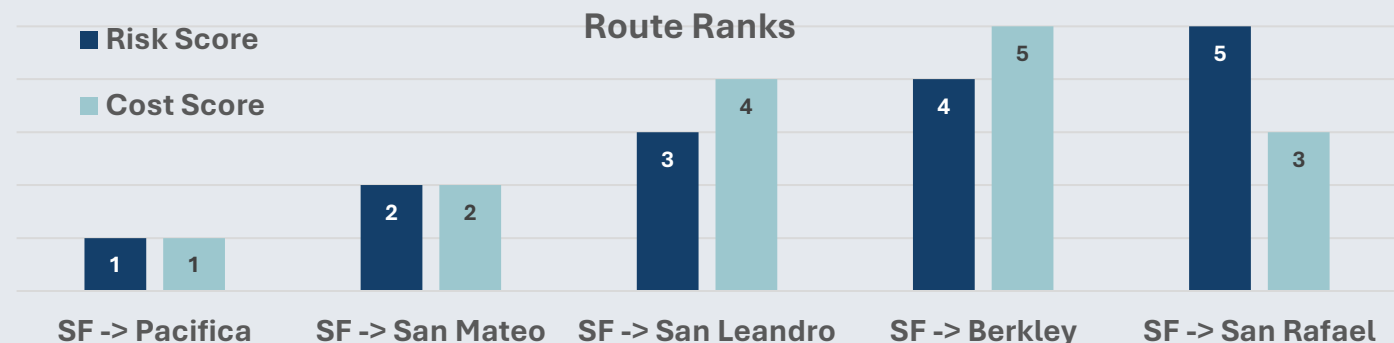
Delay Factor →

Construction type Factor * Proportion
Construction type

- Road Close: 1
- Lane Close: 0.3
- Others: 0.1

Exposure Factor →

Hour – Mile Normalization in [0 - 0.5]



Turning Analysis into Action

Findings & Actions: Roadmap Forward

Key Findings

- **Phoenix, AZ** emerges as highest risk city with 212K+ construction sites but shorter duration projects
- **COVID-19 (/Election)** caused a **shift in construction patterns from Q3-Q4 to Q1-Q2**, requiring adaptive route planning
- **SF → Pacifica** route shows **minimal construction risk**, ideal for initial deployment
- **Random Forest model** achieves over **30% accuracy** with **distance** as the primary predictor

Recommendations

- Implement **dynamic routing algorithms** for high-impact cities like **Phoenix** and **Chicago**
- Develop **seasonal deployment strategies** accounting for **Q1 construction peaks**
- **Prioritize low-risk & low-cost routes** like SF-Pacifica for initial autonomous deployment
- Enhance **model accuracy** by incorporating **road closure interaction variables**, or using a **geospatial modeling** approach

Reference Weblinks



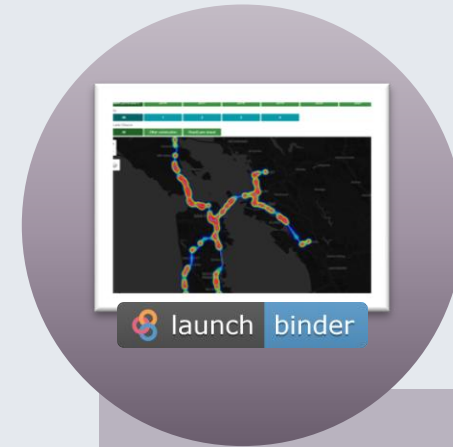
- Contain all the codes (*.py, *.ipynb)
- Python environment file
- Presentation slide deck

<https://bit.ly/4eqIn7v>



- Hosted the GitHub notebook online with binder

<https://bit.ly/44oM7l1>



- Hosted the interactive python tools for assessing construction impact on Bay Area Routes

<https://bit.ly/4l73xtV>

Let's Discuss: Your Questions - Our Path Forward

